

Final Report_Yihan Zhong

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1 Introduction

1.1 Research Question

Is smoking prevalence associated with education level among U.S. adults? Specifically, does the probability of being a current smoker decrease as educational attainment increases?

1.2 Scientific Plausibility and Context

Tobacco use remains one of the leading preventable causes of death globally, responsible for an estimated 6.4 million deaths annually (Latvala et al. 2022). In the United States, cigarette smoking is not uniformly distributed across the population: epidemiological evidence consistently demonstrates an inverse association between educational attainment and smoking prevalence, with lower-educated individuals being more likely to smoke and less likely to successfully quit (Garrett et al. 2019).

This disparity has widened over time. Analysis of National Health Interview Survey data spanning 1966 to 2018 shows that smoking disparities by education have increased across birth cohorts, with college-educated individuals exhibiting the lowest initiation probabilities and highest cessation rates across all age groups and genders (Cao et al. 2023).

Several mechanisms may explain this pattern. Higher education is associated with greater health literacy, enabling individuals to better understand the health risks of tobacco use and navigate cessation resources. Additionally, individuals with lower educational attainment face greater socioeconomic stressors and are more frequently targeted by tobacco industry marketing (Cao et al. 2023). Evidence from a pooled analysis of 18 twin cohorts further supports a causal interpretation: within pairs of twins discordant for both education and smoking, the less-educated co-twin was consistently more likely to be a current smoker, even after controlling for shared genetic and family background (Latvala et al. 2022).

Based on this evidence, we hypothesize that smoking prevalence will decrease monotonically with increasing educational attainment among U.S. adults in the 2024 BRFSS dataset.

2 Material & Methods

2.1 Data

The data used in this report come from the 2024 Behavioral Risk Factor Surveillance System (BRFSS), an annual, cross-sectional telephone survey conducted by the U.S. Centers for Disease Control and Prevention (CDC). The BRFSS collects state-level data on health behaviors, chronic conditions, and preventive health practices among U.S. adults aged 18 years and older. The 2024 dataset was downloaded from the CDC BRFSS website in XPT format.

Two variables were selected for this analysis:

- **_RFSMOK3**: A derived variable indicating current smoking status (1 = current smoker, 2 = not a current smoker). Respondents with missing or ambiguous values were excluded.
- **EDUCA**: Self-reported highest level of education completed, coded on a 6-point ordinal scale. Respondents with missing values or codes outside 1–6 were excluded.

For analysis, **EDUCA** was recoded into three ordered groups: **Low** (codes 1–3: less than high school), **Medium** (code 4: high school graduate or GED), and **High** (codes 5–6: some college or college graduate). Smoking status was converted to a binary variable (**current_smoker**: 1 = yes, 0 = no).

Table 1: Overview of Selected Variables

Variable Name	Description	Variable Type	R Class
_RFSMOK3	Current smoking status (derived)	Categorical	numeric
EDUCA	Highest education level completed	Categorical (ordinal)	numeric
current_smoker	Binary smoker indicator (1=Yes, 0=No)	Binary	numeric
edu_group	Grouped education (Low/Medium/High)	Categorical (ordinal)	factor

2.2 Statistical Analyses

The association between education level and current smoking status was examined using two complementary approaches.

Chi-square test of independence. A Pearson chi-square test was used to assess whether smoking prevalence differed significantly across education groups. The assumptions of the chi-square test require that all expected cell frequencies are ≥ 5 , which was verified prior to analysis. The effect size was quantified using Cramér’s V, where values of 0.1, 0.3, and 0.5 indicate small, medium, and large associations, respectively.

Logistic regression. To quantify the association between education and smoking while accounting for the ordinal nature of the exposure, a binary logistic regression model was fitted with `current_smoker` as the outcome and `edu_group` as the predictor, using the High education group as the reference category. Odds ratios (OR) and 95% confidence intervals (CI) are reported. A trend test was additionally conducted by treating education as a numeric ordinal variable (Low = 1, Medium = 2, High = 3) to assess whether smoking odds changed monotonically across groups.

All analyses were conducted in R. The significance threshold was set at $\alpha = 0.05$.

2.3 Code

All code and data documentation required to replicate this report are available at: <https://github.com/muxiao1019-bot/VTPEH6270-CP06>

3 Results

3.1 Visualization

3.2 Statistical Analyses

3.2.1 Chi-square Test

Table 2: Contingency Table: Education Group vs Smoking Status

	0	1
Low	19054	5608
Medium	88876	17579
High	269041	23740

Table 3: Expected Cell Frequencies (all must be ≥ 5)

	0	1
Low	21931.8	2730.2
Medium	94670.1	11784.9
High	260369.1	32411.9

All expected cell frequencies were well above 5, confirming that the chi-square test assumption was met.

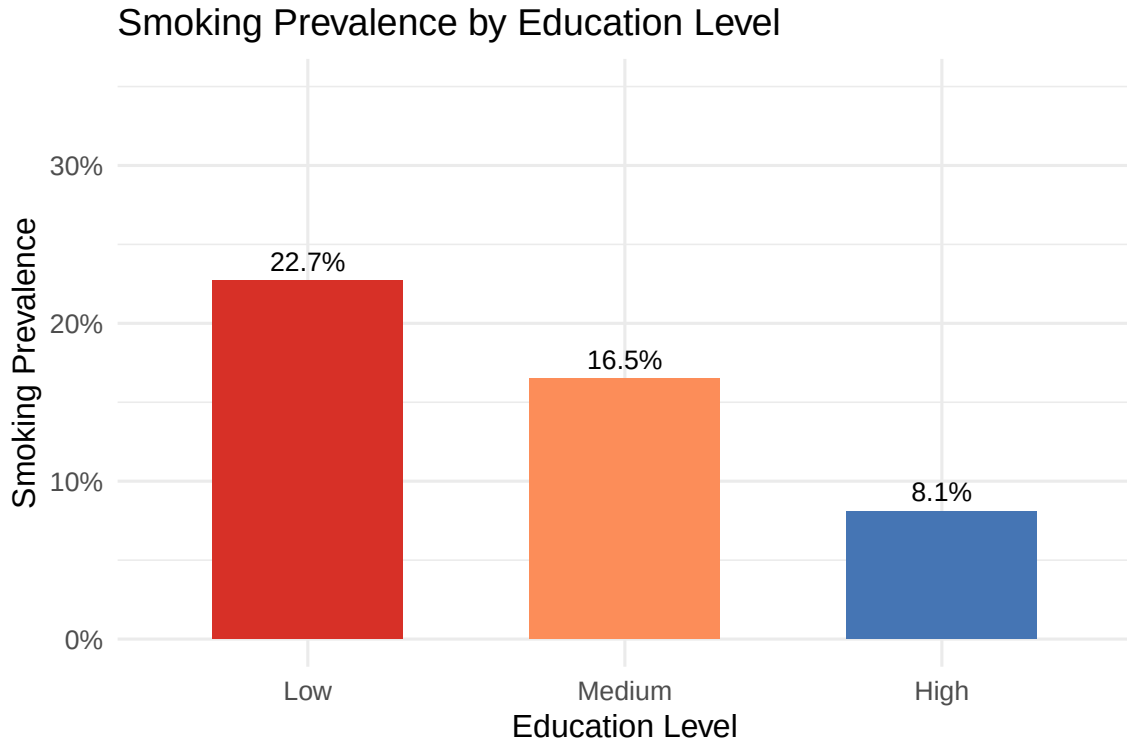


Figure 1: Smoking prevalence decreases monotonically with higher educational attainment (2024 BRFSS).

```
##
## Pearson's Chi-squared test
##
## data:  cont_table
## X-squared = 9223.4, df = 2, p-value < 2.2e-16

## Cramer's V: 0.1475
```

3.2.2 Logistic Regression

Table 4: Odds Ratios from Logistic Regression (Reference: High education)

	OR	2.5 %	97.5 %
(Intercept)	0.088	0.087	0.089
edu_groupLow	3.335	3.228	3.446
edu_groupMedium	2.242	2.195	2.289

Table 5: Trend Test: OR per Unit Increase in Education Level

	OR	2.5 %	97.5 %
(Intercept)	0.677	0.653	0.701

	OR	2.5 %	97.5 %
edu_numeric	0.513	0.506	0.520

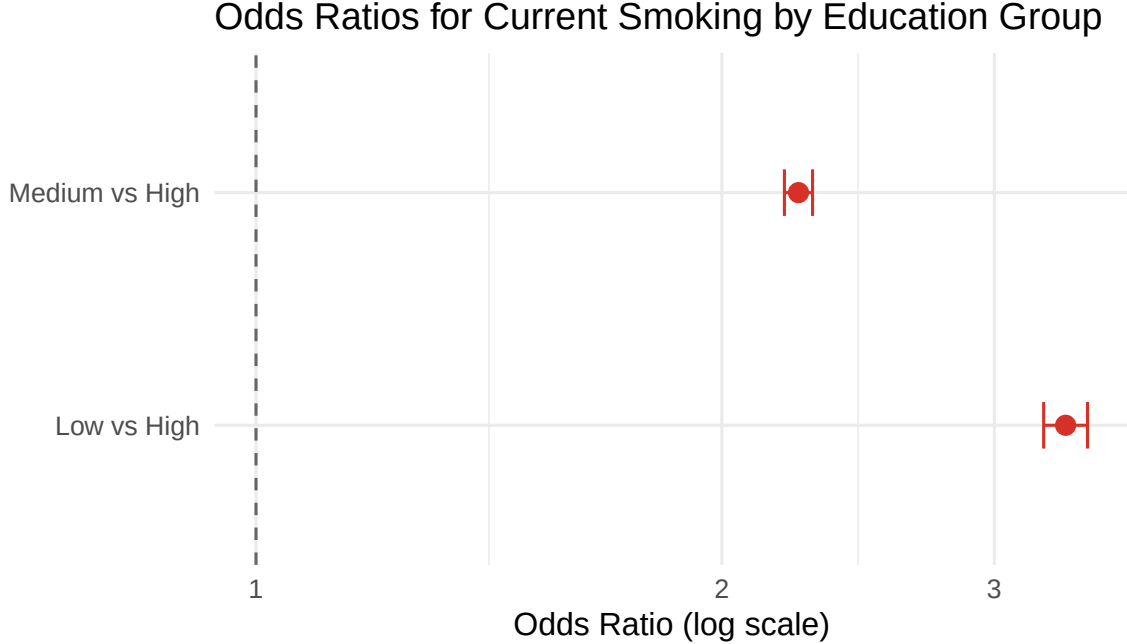


Figure 2: Odds ratios for current smoking by education group relative to High education (logistic regression).

The chi-square test revealed a statistically significant association between education level and smoking status ($\chi^2 = 9223.4$, $df = 2$, $p < 0.001$). Cramér’s $V = 0.148$, indicating a small-to-moderate effect size.

Logistic regression showed that compared to individuals with high educational attainment, those with low education had significantly higher odds of being a current smoker (OR = 3.34, 95% CI: 3.23–3.45), as did those with medium education (OR = 2.24, 95% CI: 2.20–2.29). The trend test confirmed a significant monotonic decrease in smoking odds with increasing education (OR per unit increase = 0.51, $p < 0.001$).

4 Conclusions

This analysis of the 2024 BRFSS dataset provides strong evidence of an inverse association between educational attainment and current smoking prevalence among U.S. adults. Both the chi-square test and logistic regression confirmed that individuals with lower education were significantly more likely to be current smokers, with a clear monotonic trend across the three education groups. These findings are consistent with the existing literature documenting persistent and widening educational disparities in tobacco use in the United States (Garrett et al. 2019; Cao et al. 2023).

The observed pattern aligns with proposed mechanisms linking education to smoking behavior, including greater health literacy and access to cessation resources among more educated individuals (Latvala et al. 2022). The large sample size of the BRFSS — consistent with simulations conducted in Checkpoint 04 — ensured high statistical power to detect even modest effect sizes, as reflected by the highly significant chi-square result. However, the relatively small Cramér’s V value suggests that while the association is statistically robust, education alone explains a limited proportion of the variance in smoking behavior, and other socioeconomic and contextual factors likely contribute to the observed disparities.

From a public health perspective, these findings underscore the importance of targeting tobacco control efforts toward lower-educated populations, who bear a disproportionate burden of smoking-related disease and mortality.

5 AI Use Disclosure Statement

This document was prepared with assistance from Claude (Anthropic). The tool was used to generate the initial R Markdown structure and develop the statistical analysis code, all code and interpretations were reviewed and verified by the author.

References

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